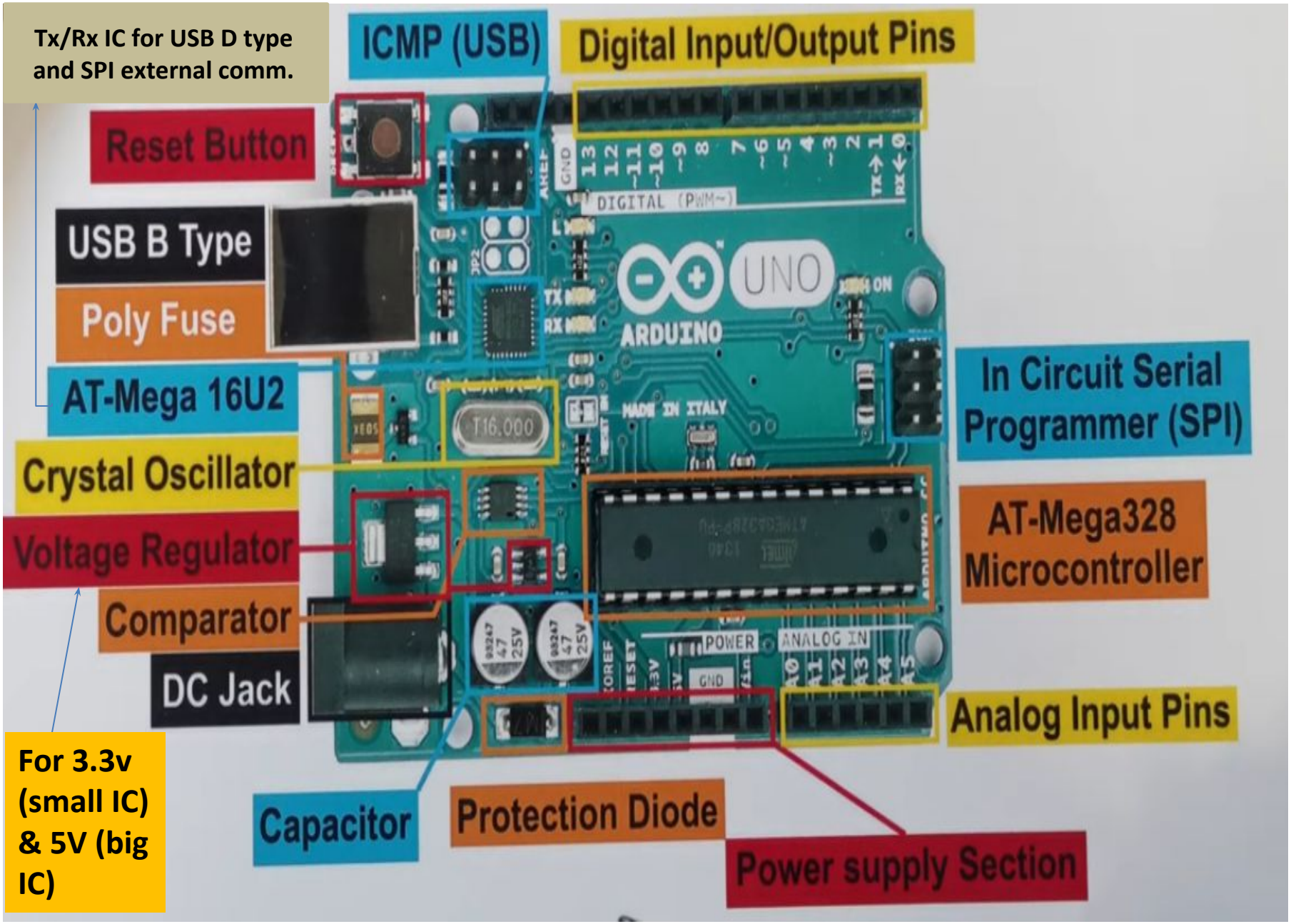
# Introduction to Arduino

- Arduino is an open-source platform used for building electronics projects.
- In its physical programmable circuit board it consists of a microcontroller and an IDE (integrated development environment) that runs on computer used to upload codes on the microcontroller as well as it can be used for data monitoring.
- The Atmel microcontroller is used in Arduino Boards.
- The Arduino was designed for designers, hobyists, hackers, newbies or anyone who is interested to create innovative projects using sensors and actuators.
- Its applications include, electronics projects, security systems, robots, home automation, IoT applications.


Tx/Rx IC for USB D type and SPI external comm.
ICMP (USB)
Digital Input/Output Pins
Reset Button
USB B Type
Poly Fuse
AT-Mega 16U2
Crystal Oscillator
Voltage Regulator
Comparator
DC Jack
For 3.3v (small IC) & 5V (big IC)
Capacitor
Protection Diode
Power supply Section
Analog Input Pins
AT-Mega328 Microcontroller
In Circuit Serial Programmer (SPI)
DIGITAL (PWM~)
UNO
ARDUINO
MADE IN ITALY
POWER
ANALOG IN
GND

# Types of Arduino

1. **Arduino Uno (R3):** it has 14 digital i/o pins (of which 6 can be used as pwm outputs), 6 analog input pins, a D type communication port for code uploading and serial monitoring.

- It uses microchip AVR (8-bit) Atmega328P microcontroller. It is having 32 kb of flash memory and 1kb EEPROM.

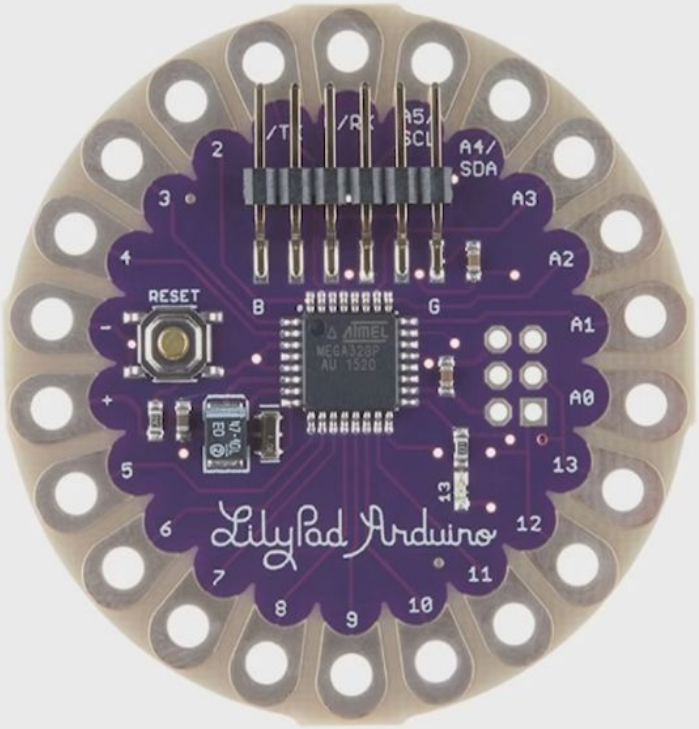
# 1. Arduino UNO

- Digital pin - 14 ( 6 PWM pin)
- analog input pin - 6
- microcontroller - Atmega 328p
- clock speed - 16 Mhz
- USB type B

# 3. Arduino NANO

- Digital pin - 14 ( 6 PWM pin)
- analog input pin - 8
- microcontroller - Atmega 328p
- clock speed - 16 Mhz
- USB mini type B

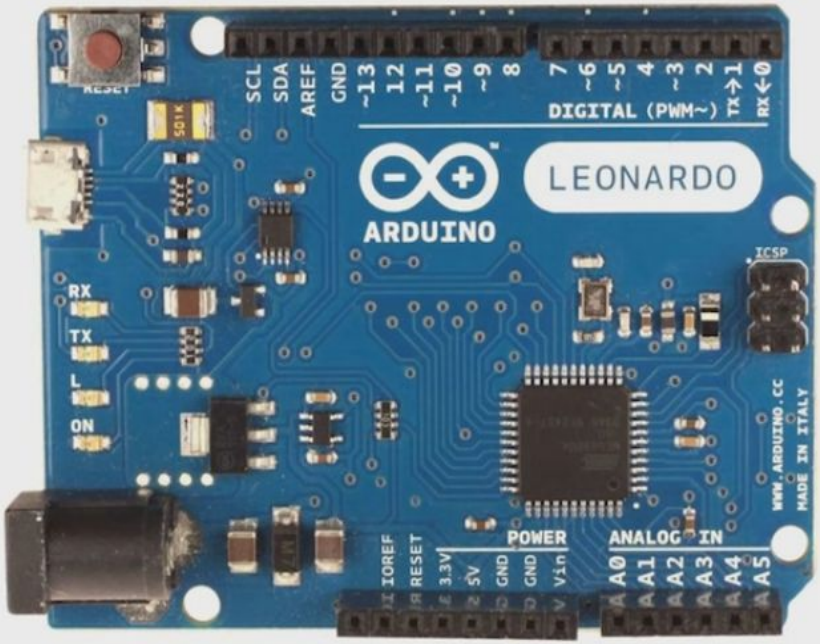
## LilyPad Arduino



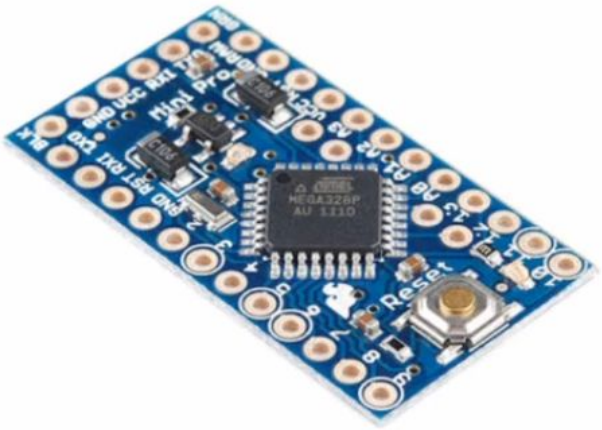
# 4. Arduino Mega

- Digital pin - 54 ( 15 PWM pin)
- analog input pin - 16
- microcontroller - Atmega 2560
- clock speed - 16 Mhz

# Arduino Leonardo



# 5. Arduino pro mini



- Digital pin - 14 ( 6 PWM pin)
- analog input pin - 6
- microcontroller - Atmega 328p

(1.) Atmega 328p - 5V - 16Mhz
(2.) Atmega 328p - 3.3V - 8Mhz

There is no USB port available for communication so either an extra Arduino will be used for communication or an FTDI connector is used.